

CS6660: MATHEMATICAL FOUNDATIONS OF DATA SCIENCE
(PROBABILITY)
PRACTICE PROBLEMS 02



TOPICS Independence, Jointly Discrete/Continuous Random Variables

1. Suppose that X and Y are jointly continuous with the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} cx(y-x)e^{-y}, & 0 \leq x \leq y < +\infty, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Determine the constant c .
(b) Show that

$$f_{X|\{Y=y\}}(x) = \begin{cases} 6x(y-x)y^{-3}, & 0 \leq x \leq y, \\ 0, & \text{otherwise,} \end{cases} \quad f_{Y|\{X=x\}}(y) = \begin{cases} (y-x)e^{x-y}, & y \geq x, \\ 0, & \text{otherwise.} \end{cases}$$

2. (a) Show that if $X \sim \text{Poisson}(\lambda)$ for some fixed $\lambda > 0$, then its PMF p_X satisfies the relation

$$p_X(k-1) \cdot p_X(k+1) \leq (p_X(k))^2 \quad \forall k \in \mathbb{N}.$$

- (b) Give an example for a PMF p_X satisfying the relation

$$p_X(k-1) \cdot p_X(k+1) = (p_X(k))^2 \quad \forall k \in \mathbb{N}.$$

3. Virat and Anushka have a date at 7 pm. Each will arrive at the meeting place with a delay that is distributed uniformly randomly between 0 minutes and 60 minutes, independent of the delay of the other. The first to arrive will wait for 15 minutes and leave if the other does not arrive within 15 minutes. Find the probability that both meet.
4. Let X be a random variable. Determine, in each case below, if the function therein can be a valid CDF of X . If not, provide at least one valid justification. For each valid CDF, compute $\mathbb{P}(\{X > 5\})$.

(a) $F_X(x) = \begin{cases} \frac{e^{-x^2}}{4}, & x < 0, \\ 1 - \frac{e^{-x^2}}{4}, & x \geq 0. \end{cases}$

(b) $F_X(x) = \begin{cases} 0, & x < 0, \\ 0.5 + e^{-x}, & 0 \leq x < 3, \\ 1, & x \geq 3. \end{cases}$

(c) $F_X(x) = \begin{cases} 0, & x < 0, \\ 0.5 + \frac{x}{20}, & 0 \leq x < 10, \\ 1, & x \geq 10. \end{cases}$

5. Let X and Y be continuous random variables with PDFs f_X and f_Y respectively. For any $\alpha \in [0, 1]$, argue that $\alpha f_X + (1 - \alpha) f_Y$ is a valid PDF. Can you think of a random variable Z whose PDF is $f_Z = \alpha f_X + (1 - \alpha) f_Y$?
6. Let X and Y be independent Poisson variables with parameters λ and μ respectively. Fix $n \in \mathbb{N}$. Determine the conditional PMF of X , conditioned on the event $\{X + Y = n\}$.
7. Suppose that two batteries are chosen simultaneously and uniformly at random from the following group of 12 batteries : 3 new, 4 used (yet working), 5 defective. You may assume that all batteries within a particular group are identical. Let X denote the number of new batteries chosen, and let Y denote the number of used batteries chosen. Determine the joint PMF of X and Y , and compute $\mathbb{P}(\{|X - Y| \leq 1\})$.